

SUPPLY CHAIN DASHBOARD

Supply Chain Dashboard

Project Documentation

Customer Segmentation Data Insights for Supply Chain Using Power BI

Prepared by: Chukwuemeka Jayfresh
Tool: Power BI Desktop | Dashboard Views: 3 | KPIs Tracked: 5

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1. Project Overview

This project empowers the company to optimize production planning and supply chain efficiency by leveraging customer data for segmentation and profiling. Using Power BI and Power Query, it drives data-driven decisions to reduce inefficiencies, align production with demand, and minimize excess inventory.

The project produced a three-page interactive Power BI dashboard covering Overview Insights, Product Insights, and Supplier Insights — enabling stakeholders across departments to monitor KPIs, filter by SKU or Supplier, and act on real-time supply chain intelligence.

2. Business Introduction

The company is a globally recognized industry leader in consumer electronics and household appliances. Established in 1995, the company has consistently pushed the boundaries of innovation, earning a reputation for cutting-edge technology and unparalleled product quality.

The company has garnered numerous industry accolades, including the prestigious Innovation Award at the International Electronics Show for its groundbreaking smart home appliances. Its diverse product portfolio — from energy-efficient smart refrigerators to state-of-the-art home entertainment systems — enhances the quality of life for millions of customers worldwide.

3. Business Problem

The company faces a significant challenge in optimizing its supply chain and production planning. The traditional approach — based primarily on historical data and market forecasts — creates the following obstacles:

Challenge	Description
Inefficient Production	Relying on historical data often results in overproduction or underproduction, causing operational inefficiencies and increased costs.
Customer Disconnection	The company lacks a deep understanding of its diverse customer base, making it challenging to align production with specific customer demands.
Inventory Management	Inefficient production planning has led to excess inventory, tying up valuable resources and impacting the company's profitability.

4. Business Objectives

The primary objectives of this project are:

- **Customer Segmentation**
 - Segment and profile the company's customer base to gain insights into their preferences, buying behaviors, and geographic distribution.
- **Data-Driven Planning**
 - Enhance production planning, inventory management, and distribution strategies based on customer segments — reducing overproduction and minimizing excess inventory.
- **Customer-Centric Approach**
 - Improve on-time deliveries, increase customer satisfaction, and align production with customer demands.

5. Tools Used

Tool	Purpose
Power Query	Data cleaning, validation, transformation, and preparation
Power Query — Applied Steps	Step-by-step data shaping: removing nulls, duplicates, type changes, and column splits
Power BI Desktop	Dashboard design, report layout, and interactive visual creation
Power BI — Visuals	Bar charts, donut charts, column charts, scatter plots, and KPI cards
Power BI — Slicers	Interactive filtering by SKU and Supplier across all dashboard pages
Power BI — DAX	Measures for Total Revenue, Avg Profit Margin, Total Products Sold, and Stock Level

6. Data Preparation & Cleaning Process

Before building the dashboard, a structured data preparation process was followed to ensure clean, reliable data for analysis. All steps were performed in Power Query within Power BI Desktop.

Step 1 — Check for Missing Values

Purpose: Identify blank or null cells across the dataset that could skew analysis results.

Step	Action	Details
1	Load data into Power Query	In Power BI Desktop, click 'Transform Data' to open the Power Query Editor with your dataset.
2	Open Column Quality view	Go to the View tab in Power Query and enable 'Column Quality'. This shows Valid, Error, and Empty % per column.
3	Identify null columns	Columns with non-zero Empty % contain missing values. Click the column header to inspect affected rows.
4	Handle missing values	Right-click the column > Replace Values. Use 'Unknown' for categorical fields or a median/mean value for numeric fields.
5	Apply and close	Click 'Close & Apply' to commit changes and reload the cleaned data into the Power BI model.

Step 2 — Find and Remove Duplicates

Purpose: Ensure each transaction and customer record is unique, preventing double-counting in aggregations.

Step	Action	Details
1	Select the query	In the Power Query Editor, select the table/query containing your supply chain data.
2	Open Remove Duplicates	Select the key column(s) (e.g., Order ID, Customer ID), then go to Home tab > Remove Rows > Remove Duplicates.
3	Choose identifying columns	To remove duplicates based on multiple columns, select all relevant columns using Ctrl + Click before applying.
4	Review row count	Check the row count in the status bar before and after to confirm duplicates were removed correctly.
5	Validate in Applied Steps	Confirm the 'Removed Duplicates' step appears in the Applied Steps pane on the right panel.

Step 3 — Data Transformation

Purpose: Convert raw timestamp and text data into properly formatted fields for time-series and categorical analysis.

Step	Action	Details
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1	Split date-time columns	Select the timestamp column, go to Transform tab > Split Column > By Delimiter, using space to separate date and time.
2	Change date column type	Click the data type icon on the date column header and select 'Date' to convert it from text to a proper date type.
3	Change time column type	Similarly, set the time column type to 'Time' using the data type icon.
4	Rename columns	Double-click column headers to rename them clearly (e.g., 'Order Date', 'Order Time') for report readability.
5	Encode categorical fields	Use Transform > Replace Values or Add Column > Conditional Column to map and standardize text categories.

Step 4 — Verify Data Types

Purpose: Confirm all columns have the correct data type for accurate calculations, pivot tables, and chart generation.

Step	Action	Details
1	Review column type icons	Each column header in Power Query shows a type icon (ABC for text, 123 for whole number, calendar for date). Verify all are correct.
2	Fix text-stored numbers	Click the type icon on affected columns and select 'Whole Number' or 'Decimal Number' to convert.
3	Validate date columns	Confirm date columns show the calendar icon and display dates in the preview without errors.
4	Check for error rows	Enable Column Quality view (View tab) and click 'Errors' on any column showing error %. Inspect and resolve each error row.
5	Final audit via Column Profile	Enable 'Column Profile' in the View tab to see value distribution, min/max, and distinct counts for a full data audit.

7. Dashboard Walkthrough

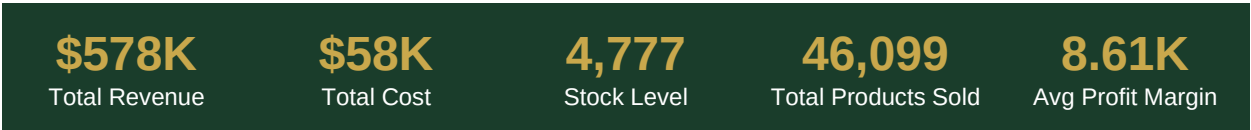
The Supply Chain Dashboard is organized into three distinct pages, accessible via the left-hand navigation panel in Power BI. Each page targets a different layer of the supply chain — from high-level financials to product-level operations and supplier performance.

Interactive slicers for SKU and Supplier appear on every page, allowing users to filter all visuals simultaneously for a specific product line or vendor.

7.1 Overview Insights

The Overview page provides a high-level snapshot of the business's financial and operational health.

KPI Summary Cards



Charts on This Page

Chart	Type	Key Finding
Total Revenue by Customer (Gender)	Donut Chart	Male customers drive 27.96% of revenue; Unknown is the largest segment at 29.97%.
Total Revenue by Customer (Product)	Donut Chart	Haircare leads at 41.01%, followed by Skincare (37.08%) and Cosmetics (21.91%).
Total Revenue by Product Type	Bar Chart	Skincare tops revenue at \$0.24M, followed by Haircare (\$0.17M) and Cosmetics (\$0.16M).
Total Revenue by SKU	Bar Chart	SKU51 is the highest-revenue SKU at \$9.9K, closely followed by SKU38 and SKU31.
Total Revenue by Shipping Carrier	Column Chart	Carrier B leads with \$0.25M, ahead of Carrier C (\$0.18M) and Carrier A (\$0.14M).
Avg Profit Margin by Supplier	Table	Supplier 1 has the highest avg margin at 2,272.28, while Supplier 3 has the lowest at 1,366.17.

7.2 Product Insights

The Product Insights page focuses on product-level operations, comparing order quantities, stock levels, and manufacturing lead times across all SKUs.

KPI Summary Cards



Charts on This Page

Chart	Type	Key Finding
Order Quantities by SKU	Column Chart	Order quantities are relatively consistent across SKUs, indicating balanced demand distribution.
Stock Levels by SKU	Column Chart	Stock levels vary across SKUs; high-stock items may signal overproduction risk.
Manufacturing Lead Time by SKU	Stacked Bar	Lead time and Mfg lead time are broken down per SKU, identifying production bottlenecks.
Revenue by Customer (Product Type)	Donut Chart	Skincare leads at 39.96%, followed by Cosmetics (33.68%) and Haircare (26.36%).
Price vs Products Sold by Product Type	Scatter Plot	Shows price sensitivity across Skincare, Haircare, and Cosmetics product types.

7.3 Supplier Insights

The Supplier Insights page evaluates supplier performance, transportation modes, defect rates, and cost structures to guide sourcing decisions.

Charts on This Page

Chart	Type	Key Finding
Total Customer by Transportation	Donut Chart	Road is the dominant mode at 30.52%, followed by Sea (28.71%), Rail (27.25%), and Air (13.52%).

Avg Defect % by Transportation	Donut Chart	Road also leads defect rates at 33.38%, suggesting quality trade-offs in this transport mode.
Supplier Costs Analysis	Stacked Bar	Supplier 1 has the highest total costs (Costs + Mfg Costs + Shipping Costs); Supplier 3 is lowest.
Supplier Stock Levels & Location	Table	All five suppliers are based in Bangalore or Chennai; stock levels range from 89 to 202 units.
Defect Rate vs Profit Margin by Supplier	Scatter Plot	Identifies which suppliers have high defect rates relative to their profit margin contribution.

8. Key Insights & Recommendations

Based on the dashboard analysis, the following insights and recommended actions are identified:

#	Insight	Recommended Action
1	Skincare is the top revenue-generating product type at \$0.24M.	Prioritize Skincare SKUs in production planning and inventory allocation.
2	Male customers and Unknown segments each contribute ~28-30% of revenue.	Investigate 'Unknown' customer data gaps and design targeted campaigns for male demographics.
3	Carrier B handles the most revenue at \$0.25M.	Negotiate preferred rates with Carrier B and evaluate capacity scalability.
4	Supplier 1 has the highest profit margin (2,272) but also the highest costs.	Conduct a cost-margin audit on Supplier 1 to identify optimization opportunities.
5	Road transportation has the highest defect rate at 33.38%.	Review road-transport packaging and carrier quality controls to reduce defect rates.
6	Stock levels vary significantly across SKUs.	Implement SKU-level reorder points to address high-stock overproduction and low-stock risk.
7	All suppliers are concentrated in Bangalore and Chennai.	Explore geographic diversification of suppliers to reduce regional supply chain risk.
8	SKU51, SKU38, and SKU31 are the highest-revenue SKUs.	Ensure these SKUs are consistently stocked and given priority in Mfg scheduling.